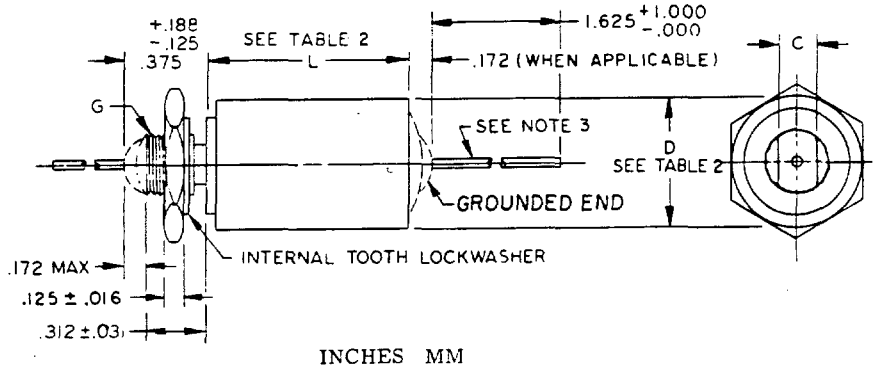
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, PLASTIC (OR PAPER-PLASTIC) DIELECTRIC, AXIAL-WIRE TERMINAL, THREADED-STUD RETAINER, TUBULAR (UNINSULATED) (HERMETICALLY SEALED IN METAL CASES), ESTABLISHED RELIABILITY, STYLE CQR43

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-19978.

INACTIVE FOR NEW EQUIPMENT DESIGN AFTER DATE OF THIS SPECIFICATION SHEET



INCHES	MM
.001	.03
.004	.10
.005	.13
.032	.81
.125	3.18
.172	4.37
.250	6.35
.375	9.53
1.625	41.28

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Leads shall be of tinned solidwire , .032 +.004, -.001 (No. 20 AWG).
4. Mounting nuts and lockwashers shall be supplied.

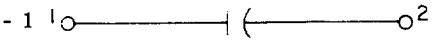
FIGURE 1. Style CQR43 capacitors.

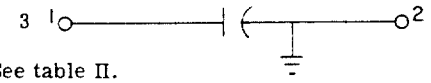
FSC 5910

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REQUIREMENTS:

Design and construction:
Dimensions and configuration - See figure 1 and table II.
Case material - Nonmagnetic (screwneck and end seal may be of magnetic material).
Dielectric material - Paper and polyethelene terephthalate (K).
Terminals - Axial-wire lead (see figure 1).

Circuit diagram - 1 

3 

Rated voltage - See table II.
Rated temperature - -65° to +125° C.
Capacitance (Cap.) (nom) - See table II.
Capacitance tolerance - K, ±10 percent.
Dissipation factor (DF) (max) - 1.0 percent.
Failure rate level - In accordance with MIL-C-19978.
Reliability - In accordance with MIL-C-19978.

Burn-in: In accordance with MIL-C-19978

Radiographic inspection: In accordance with MIL-C-19978

Seal: Method 112 of MIL-STD-202, test condition letter - A.

Barometric pressure: In accordance with MIL-C-19978.
Test points:
Circuit diagram 1 - Between terminals and case.
Circuit diagram 3 - Between ngrounded terminal and case.

Insulation resistance (IR):
Terminal to terminal - See table I.

TABLE I. Terminal-to-terminal insulation resistance.

Capacitance rating	Minimum insulation resistance
Characteristic K	
	At 25° C
0 to 0.6 microfarad	25,000 megohms
Greater than 0.6 microfarad	15,000 megohm-microfarads 1/
	At 125° C
0 to 0.08 microfarad	250 megohms
Greater than 0.08 microfarad	20 megohm-microfarads 1/

1/ Product obtained by multiplying the capacitance in microfarads by the insulation resistance in megohms.

Terminal to case - Greater than 10,000 megohms.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B, with the following exception:
Direction and duration of motion - 4 hours in each of two mutually perpendicular directions (total of 8 hours), one parallel and the other perpendicular to the cylindrical axis.

Salt spray: In accordance with MIL-C-19978.

Solderability: In accordance with MIL-C-19978.

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TABLE II. Style CQR43 capacitors.

Part number 1/	DC voltage rating	Capacitance rating	Failure rate level	Case dimensions			
				Characteristic K			
				Circuit 1		Circuit 3	
				L±.031	$D^{+.015}_{-.005}$	L±.031	$D^{+.015}_{-.005}$
	Volts	μF		Inches	Inches	Inches	Inches
CQR43A-KB822K3-	100	.0082	L, M, P	.771	.400	.708	.400
CQR43A-KB103K3-	100	.010		.771	.400	.708	.400
CQR43A-KB123K3-	100	.012		.771	.400	.708	.400
CQR43A-KB153K3-	100	.015		.771	.400	.708	.400
CQR43A-KB393K3-	100	.039		.896	.400	.833	.400
CQR43A-KB473K3-	100	.047		.896	.400	.833	.400
CQR43A-KB823K3-	100	.082		.896	.400	.833	.400
CQR43A-KB104K3-	100	.10		.896	.400	.833	.400
CQR43A-KB124K3-	100	.12		1.146	.400	1.083	.400
CQR43A-KB154K3-	100	.15		1.146	.400	1.083	.400
CQR43A-KB184K3-	100	.18		1.396	.400	1.333	.400
CQR43A-KB224K3-	100	.22		1.396	.400	1.333	.400
CQR43A-KB274K3-	100	.27		1.146	.562	1.083	.562
CQR43A-KB334K3-	100	.33		1.146	.562	1.083	.562
CQR43A-KB394K3-	100	.39		1.406	.562	1.344	.562
CQR43A-KB474K3-	100	.47		1.406	.562	1.344	.562
CQR43A-KB564K3-	100	.56		1.656	.562	1.594	.562
CQR43A-KB684K3-	100	.68		1.656	.562	1.594	.562
CQR43A-KB824K3-	100	.82		1.656	.670	1.594	.670
CQR43A-KB105K3-	100	1.0		1.656	.670	1.594	.670
CQR43A-KC392K3-	200	.0039		.771	.400	.708	.400
CQR43A-KC472K3-	200	.0047		.771	.400	.708	.400
CQR43A-KC562K3-	200	.0056		.771	.400	.708	.400
CQR43A-KC682K3-	200	.0068		.771	.400	.708	.400
CQR43A-KC183K3-	200	.018		.896	.400	.833	.400
CQR43A-KC223K3-	200	.022		.896	.400	.833	.400
CQR43A-KC273K3-	200	.027		.896	.400	.833	.400
CQR43A-KC333K3-	200	.033		.896	.400	.833	.400
CQR43A-KC393K3-	200	.039		.896	.400	.833	.400
CQR43A-KC473K3-	200	.047		.896	.400	.833	.400
CQR43A-KC563K3-	200	.056		.896	.400	.833	.400
CQR43A-KC683K3-	200	.068		.896	.400	.833	.400
CQR43A-KC823K3-	200	.082		1.146	.400	1.083	.400
CQR43A-KC104K3-	200	.10		1.146	.400	1.083	.400
CQR43A-KC124K3-	200	.12		1.396	.400	1.333	.400
CQR43A-KC154K3-	200	.15		1.396	.400	1.333	.400
CQR43A-KC184K3-	200	.18		1.146	.562	1.083	.562
CQR43A-KC224K3-	200	.22		1.146	.562	1.083	.562
CQR43A-KC274K3-	200	.27		1.406	.562	1.344	.562
CQR43A-KC334K3-	200	.33		1.406	.562	1.344	.562
CQR43A-KC394K3-	200	.39		1.656	.562	1.594	.562
CQR43A-KC474K3-	200	.47		1.656	.562	1.594	.562
CQR43A-KC564K3-	200	.56		1.656	.670	1.594	.670
CQR43A-KC684K3-	200	.68		1.656	.670	1.594	.670
CQR43A-KC824K3-	200	.82		2.156	.750	2.094	.750
CQR43A-KC105K3-	200	1.0	L, M, P	2.156	.750	2.094	.750

See footnote at end of table.

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TABLE II. Style CQR43 capacitors. -Continued

Part number 1/	DC voltage rating	Capacitance rating	Failure rate level	Case dimensions			
				Characteristic K			
				Circuit 1		Circuit 3	
				L±.031	D +.015 -.005	L±.031	D +.015 -.005
	Volts	µF		Inches	Inches	Inches	Inches
CQR43A-KE272K3-	400	.0027	L, M, P	.771	.400	.708	.400
CQR43A-KE332K3-	400	.0033		.771	.400	.708	.400
CQR43A-KE123K3-	400	.012		.896	.400	.833	.400
CQR43A-KE153K3-	400	.015		.896	.400	.833	.400
CQR43A-KE273K3-	400	.027		.896	.400	.833	.400
CQR43A-KE333K3-	400	.033		.896	.400	.833	.400
CQR43A-KE393K3-	400	.039		1.146	.400	1.083	.400
CQR43A-KE473K3-	400	.047		1.146	.400	1.083	.400
CQR43A-KE563K3-	400	.056		1.396	.400	1.333	.400
CQR43A-KE683K3-	400	.068		1.396	.400	1.333	.400
CQR43A-KE823K3-	400	.082		1.146	.562	1.083	.562
CQR43A-KE104K3-	400	.10		1.146	.562	1.083	.562
CQR43A-KE124K3-	400	.12		1.406	.562	1.344	.562
CQR43A-KE154K3-	400	.15		1.406	.562	1.344	.562
CQR43A-KE184K3-	400	.18		1.656	.562	1.594	.562
CQR43A-KE224K3-	400	.22		1.656	.562	1.594	.562
CQR43A-KE274K3-	400	.27		1.656	.670	1.594	.670
CQR43A-KE334K3-	400	.33		1.656	.670	1.594	.670
CQR43A-KE394K3-	400	.39		2.156	.750	2.094	.750
CQR43A-KE474K3-	400	.47		2.156	.750	2.094	.750
CQR43A-KF102K3-	600	.0010		.771	.400	.708	.400
CQR43A-KF122K3-	600	.0012		.771	.400	.708	.400
CQR43A-KF152K3-	600	.0015		.771	.400	.708	.400
CQR43A-KF182K3-	600	.0018		.771	.400	.708	.400
CQR43A-KF222K3-	600	.0022		.771	.400	.708	.400
CQR43A-KF272K3-	600	.0027		.896	.400	.833	.400
CQR43A-KF332K3-	600	.0033		.896	.400	.833	.400
CQR43A-KF392K3-	600	.0039		.896	.400	.833	.400
CQR43A-KF472K3-	600	.0047		.896	.400	.833	.400
CQR43A-KF562K3-	600	.0056		.896	.400	.833	.400
CQR43A-KF682K3-	600	.0068		.896	.400	.833	.400
CQR43A-KF822K3-	600	.0082		.896	.400	.833	.400
CQR43A-KF103K3-	600	.010		.896	.400	.833	.400
CQR43A-KF123K3-	600	.012		.896	.400	.833	.400
CQR43A-KF153K3-	600	.015		.896	.400	.833	.400
CQR43A-KF183K3-	600	.018		.896	.400	.833	.400
CQR43A-KF223K3-	600	.022		.896	.400	.833	.400
CQR43A-KF273K3-	600	.027		1.146	.400	1.083	.400
CQR43A-KF333K3-	600	.033		1.146	.400	1.083	.400
CQR43A-KF393K3-	600	.039		1.396	.400	1.333	.400
CQR43A-KF473K3-	600	.047		1.396	.400	1.333	.400
CQR43A-KF563K3-	600	.056		1.146	.562	1.083	.562
CQR43A-KF683K3-	600	.068		1.146	.562	1.083	.562
CQR43A-KF823K3-	600	.082		1.406	.562	1.344	.562
CQR43A-KF104K3-	600	.10	L, M, P	1.406	.562	1.344	.562

See footnote at end of table.

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TABLE II. Style CQR43 capacitors. -Continued

Part number 1/	DC voltage rating	Capacitance rating	Failure rate level	Case dimensions			
				Characteristic K			
				Circuit 1		Circuit 3	
				L±.031	$D^{+.015}_{-.005}$	L±.031	$D^{+.015}_{-.005}$
	Volts	μF		Inches	Inches	Inches	Inches
CQR43A-KF124K3-	600	.12	L, M, P	1.656	.562	1.594	.562
CQR43A-KF154K3-	600	.15		1.656	.562	1.594	.562
CQR43A-KF184K3-	600	.18		1.656	.670	1.594	.670
CQR43A-KF224K3-	600	.22		1.656	.670	1.594	.670
CQR43A-KF274K3-	600	.27		2.156	.750	2.094	.750
CQR43A-KF334K3-	600	.33		2.156	.750	2.094	.750
CQR43A-KF394K3-	600	.39		2.406	.750	2.344	.750
CQR43A-KF474K3-	600	.47	L, M, P	2.406	.750	2.344	.750

1/ Complete part number shall include additional symbols to indicate circuit and failure rate level, as applicable.

TABLE III. Millimeter to equivalent decimal inches.

INCHES	MM	INCHES	MM
.005	.13	1.083	27.51
.015	.38	1.146	29.11
.031	.79	1.333	33.86
.068	1.73	1.344	34.14
.400	10.16	1.396	35.46
.562	14.27	1.406	35.71
.670	17.02	1.594	40.49
.708	17.98	1.656	42.06
.750	19.05	2.094	53.19
.771	19.58	2.156	54.76
.833	21.16	2.344	59.54
.896	22.76	2.406	61.11

Terminal strength: Method 211 of MIL-STD-202, test condition letter - D.

Stability at low and high temperature:
Low temperature:
Test temperature - $-65^{\circ}\ ^{+0^{\circ}}_{-5^{\circ}}$ C.
Capacitance change (max) - -10 percent.
High temperature:
Test temperature - $+125^{\circ}\ ^{+5^{\circ}}_{-0^{\circ}}$ C.
Capacitance change (max) - +10 percent.

Life:
Capacitance change (max) - ±5 percent.

Resistance to soldering heat: In accordance with MIL-C-19978.

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Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MI, MU
Navy - AS
Air Force - 11, 17
DSA - ES

User activities:
Army - ME, AT, AV
Navy - MC, AS, OS, SH
Air Force -

Preparing activity:
Air Force - 80

Agent:
DSA - ES

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